Short-Term Collaboration — Stretch 4

From Localisation to a Millimetre Grasp over a Shared Link: Scenario and Literature

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1. The Scenario

One Stretch 4 mobile manipulator picks a small everyday object (a cup, a pen, a remote control) from a table, a shelf or an open drawer, starting from elsewhere in the building. It carries a base lidar and IMU, a head RGB-D camera on a pan–tilt mount, a wrist RGB-D camera beside the gripper, joint encoders and gripper effort sensing. Its arm is Cartesian (lift, telescoping arm, wrist) with no lateral joint, so sideways correction of the gripper must come from the base. The heavy models (open-vocabulary detection, grasp proposal, verification) run on a server reached over the building's shared Wi-Fi; the robot's own computer runs localisation and the low-level controllers.

The task spans building scale to millimetres. The robot localises in a prebuilt map, chooses where to stand, finds the object with the head camera, brings the arm close and finally places the fingers around the object within a few millimetres. Each stage hands the target to the next with an error, and the errors add up. The link adds one more: whatever the server returns describes the scene at the moment the image was captured, so while the image travels, waits in the server's queue, is processed and the answer returns, the robot (and possibly the object) keeps moving. A delay d at a relative speed v becomes a position error of about v·d at the gripper, in the same budget as the sensor errors.

The problem the robot faces is the following.

> *Given an error at every link from map to fingertip, including the error the link's delay creates, which stage should run on the robot and which on the server, what should be sent and when, and when should control pass from the head camera to the wrist camera, so that the gripper closes within the object's tolerance on the first attempt?*

Four properties make the problem specific.

1. Errors compound along the chain, and the link is one of them. Localisation, base execution, head-camera depth, pan–tilt calibration, arm deflection, wrist calibration and the delay of every offloaded answer each add an error at the gripper.
2. Tolerance shrinks stage by stage, from about 10 cm at the table to a few millimetres at closure, while the permissible delay shrinks with it: a stage whose tolerance is δ at relative speed v can tolerate at most δ/v of delay.
3. Where a stage runs is a choice. Offloading buys a stronger model at the price of delay and of bytes on a shared link; running on the robot buys immediacy at the price of a weaker model. The right choice changes from stage to stage.
4. The Cartesian arm needs the base. Lateral and yaw corrections at the gripper are base motions, so the final alignment is a base–arm loop whose rate a Wi-Fi round trip cannot sustain.

The quantity. The position error of the gripper relative to the object at closure, decomposed into the contribution of each stage and of the link delay (v·d per offloaded answer), measured against fiducials; together with first-attempt success, time per stage and bytes per stage.

Table 1. The stages of a pick: tolerance, delay budget, where each runs and what crosses the link.

Stage	Must be within	Delay budget at its speed	Runs on	Sent over the link
1 Localise and drive	5–10 cm	≈ 0.2–0.5 s at 0.3 m/s	robot: lidar–inertial localisation, base planner; server: semantic goal once	a low-resolution frame on request
2 Find the object	2–3 cm	≈ 0.3 s when the base is stopped, near zero while moving	server: detection and grasp proposal	one full-resolution crop of the target region
3 Pre-grasp	1–2 cm	≈ 0.1–0.2 s at 0.1 m/s arm speed	robot: arm on the stage 2 estimate; server: optional refinement	changed pixels in the target box, if refined
4 Align	2–5 mm	≈ 50 ms at 5 cm/s, below a Wi-Fi round trip under load	robot only: wrist-camera servo on base yaw and arm extension	nothing
5 Close and verify	—	≈ 0.3 s	robot: effort; server: verification	events; one crop after the lift

Delay budget = tolerance / relative speed. Values are starting points to be replaced by measurement; the loaded round trip measured on the group's own testbed was 25–33 ms, rising to 60–109 ms on a busier pass, before any server queue or inference time.

2. The Communication Design

The design has six parts. Each removes a different share of the error that the link adds at the gripper, and each is switched on in turn in the experiment of Table 2.

2.1 Motion-compensated answers. Every image goes up with its capture time and the robot's pose at that instant (base pose from localisation, joint positions from the encoders). The server answers in the frame of that capture. On arrival, the robot transforms the answer by its own motion since the capture, using odometry and forward kinematics. The robot's own motion during the delay is thereby cancelled; what remains is the motion of the scene, close to zero for an object resting on a table. This converts most of the v·d term to zero without changing the link.

2.2 Queue discipline. Under load most of the delay is queueing, not transmission. The robot publishes with best-effort, keep-last-one quality of service so that only the newest image ever waits; each request carries its stage and deadline; the server drops a request whose stage has ended and serves the rest earliest-deadline-first rather than in arrival order.

2.3 Region, change and timing of the uplink. Only the region the stage needs is sent, and within it only what changed since the server's last copy (an RGB difference and a depth difference above about 2 cm). Resolution and depth precision are set per stage. The image for the next stage is requested ahead of time, from the pose the planned motion will reach, so that its answer arrives as that stage begins rather than after.

2.4 Anytime answers. The server first returns a fast, coarse answer (a box from a light model) and then the refined one (the full grasp). The robot starts on the first and corrects on the second, so that under load a usable answer still arrives within the stage's delay budget.

2.5 Speed that follows the link. The robot measures the end-to-end delay continuously and limits its speed so that v·d stays within the current stage's tolerance: fast on an idle link, slow only where the tolerance is tight and the link is busy. Motion compensation (2.1) removes the delay the robot can predict; speed adaptation absorbs the delay it cannot.

2.6 Link configuration. Power saving off on the robot's Wi-Fi interface, the 5 GHz band pinned, request and answer traffic marked for the voice or video access category (WMM, via DSCP), and a datagram transport for images instead of a stream transport. These are configuration changes, measured like the others.

Where the novelty lies. Each part has precedent. What is new is that all of them are scored by one quantity, the gripper error at closure, and that compensation (2.1) and speed adaptation (2.5) are combined: the robot cancels the delay it can predict and slows for the delay it cannot. Offloading policies decide whether to offload and video-analytics systems decide what to send; neither has an arm whose accuracy depends on the delay.

Table 2. The experiment: parts switched on cumulatively, each under the same delay sweep.

Step	Configuration
1	Naive: every frame streamed, the latest answer acted upon
2	+ motion-compensated answers (2.1)
3	+ queue discipline (2.2)
4	+ region and change uplink and early requests (2.3)
5	+ anytime answers (2.4)
6	+ speed that follows the link (2.5)
—	each step with and without the link configuration (2.6)

Delay sweep with tc: 0, 50, 100, 200 and 500 ms, each with and without jitter and with and without a background stream. Measured per pick: gripper error at closure and its decomposition, first-attempt success, time per pick, bytes per pick. The main figure is gripper error against delay, one curve per step.

What is measured. Stamps on one clock (chrony) at capture, send, server receive, after inference and at the robot's command; AprilTags on gripper and object for the error at the end of each stage. Per pick: the error per stage and the share attributable to delay; first-attempt success; time and bytes per stage. Two reference configurations frame the comparison: everything on the server except low-level control, and everything on the robot with lighter models.

What it delivers. A measured map-to-fingertip error budget in which the link's delay is a term like any sensor's; the reduction of that term contributed by each part of the design; and a rule for speed and offloading given the measured delay.

3. The Literature by Category

Tags: [F] read in full in this project; [V] venue or headline result checked against a listing or the paper's own page; [S] abstract, summary or memory only, to be checked before citation.

3.1 Cloud and fog robotics: whether to offload

Owns the decision to run a computation remotely and the infrastructure for it. Scores average task performance or latency; does not turn delay into an error at the gripper or let the split change with the motion stage.

Table 3. Cloud and fog robotics.

Paper	Venue	Note
Kehoe, Patil, Abbeel, Goldberg, a survey of research on cloud robotics and automation	IEEE T-ASE 2015 [S]	the survey
Chinchali et al., network offloading policies for cloud robotics	RSS 2019; Auton. Robots 2021 [V]	when to offload a sensing query, by deep RL; 1.3–2.3 times better task performance than benchmark policies
FogROS2 (Ichnowski et al.)	arXiv 2205.09778 [S]	ROS 2 platform for cloud and fog robotics; video compression cuts image round-trip latency by 97 %
FogROS2-LS	2024 [S]	selects the server that keeps latency reliable; offloads estimators and controllers
Nakanoya et al., task-relevant representation learning for networked robotic perception	RSS 2021 [S]	learned task-relevant compression for robot uplink; record not found in this session

3.2 Server-driven and adaptive video analytics: what to send

Owns how a camera stream is filtered, cropped and encoded for a remote model under a bandwidth limit. Has no actuator: accuracy is per frame, not at a commit.

Table 4. Server-driven and adaptive video analytics.

Paper	Venue	Note
DDS, server-driven video streaming for deep learning inference (Du et al.)	SIGCOMM 2020 [V]	the server's model asks for regions to re-send at high quality; up to 59 % less bandwidth at higher accuracy
Reducto (Li et al.)	SIGCOMM 2020 [S]	on-camera frame filtering by feature difference
Glimpse (Chen et al.)	SenSys 2015 [S]	track on the device, offload only when needed
Liu et al., edge-assisted real-time object detection for mobile AR	MobiCom 2019 [S]	region-of-interest encoding and local tracking between server results

3.3 Split computing and edge inference: where to cut

Owns partitioning a model between device and server. One model, one cut, no motion.

Table 5. Split computing.

Paper	Venue	Note
Neurosurgeon (Kang et al.)	ASPLOS 2017 [S]	layer-level partitioning by latency and energy
Matsubara, Levorato, Restuccia, split computing and early exiting	ACM CSUR 2022 [S]	the survey

3.4 Base localisation

Owns where the base is. Reports trajectory error, not the error left at the gripper; runs on the robot.

Table 6. Base localisation.

Paper	Venue	Note
2Fast-2Lamaa (Le Gentil, Falque, Lisus, Barfoot)	IJRR 2026, arXiv 2410.05433 [F]	lidar–inertial localisation against a prebuilt map at under 16 % of a laptop CPU; evaluated on 3-D lidars only
FAST-LIO2 (Xu et al.)	T-RO 2022 [S]	iterated Kalman-filter lidar–inertial odometry
Thrun, Burgard, Fox, Probabilistic Robotics	MIT Press 2005	Monte Carlo localisation on a planar lidar, the Stretch default

3.5 Grasp detection and visual servoing

Owns where to grasp and the closed loop that gets there. Assumes the model runs where the camera is, or that the link is ideal.

Table 7. Grasp detection and visual servoing.

Paper	Venue	Note
Contact-GraspNet (Sundermeyer et al.)	ICRA 2021 [V]	6-DoF parallel-jaw grasps from a depth image
AnyGrasp (Fang et al.)	T-RO 2023 [V]	grasp perception in space and time; 93.3 % bin-clearing success on unseen objects
GG-CNN (Morrison, Corke, Leitner)	RSS 2018 [S]	fast grasp network run in a closed loop
Chaumette, Hutchinson, visual servo control I and II	IEEE RAM 2006, 2007	the tutorial
Precise mobile manipulation of small everyday objects	arXiv 2502.13964, 2025 [S]	eye-in-hand servoing guided by vision foundation models on a mobile manipulator

3.6 Mobile manipulation systems on Stretch

Own the whole pipeline in real homes, typically with a GPU workstation; report overall success, not where in the chain or in the link failures arise.

Table 8. Systems on Stretch.

Paper	Venue	Note
Kemp, Edsinger, Clever, Matulevich, The Design of Stretch	ICRA 2022 [V]	the platform
OK-Robot (Liu et al.)	arXiv 2401.12202, 2024 [V]	modular zero-shot pick-and-drop with models on a remote workstation; 58.5 % in 10 homes, 82.4 % in less cluttered rooms
Dobb-E (Shafiullah et al.)	arXiv 2311.16098, 2023 [V]	imitation from a hand-held tool; 81 % over 109 tasks in 10 homes
HomeRobot (Yenamandra et al.)	CoRL 2023 [S]	open-vocabulary mobile manipulation benchmark on Stretch

3.7 Timeliness

Table 9. Timeliness.

Paper	Venue	Note
Maatouk, Kriouile, Assaad, Ephremides, age of incorrect information	ToN 2020	freshness that counts only while the receiver is wrong
Sun, Polyanskiy, Uysal	T-IT 2020 [V]	signal-aware sampling over a random-delay channel

3.8 Delay compensation, networked control and the link layer

Owns the theory of acting on delayed information and the mechanisms for prioritising traffic. Not applied to a mobile manipulator whose grasp error is the score.

Table 10. Delay compensation, networked control and the link layer.

Paper	Venue	Note
Smith, closer control of loops with dead time	Chem. Eng. Prog. 1957	the Smith predictor: compensate a known delay with a model of the plant
Hespanha, Naghshtabrizi, Xu, a survey of recent results in networked control systems	Proc. IEEE 2007	control over delayed, lossy networks
Bejczy, Kim, Venema, the phantom robot: predictive displays for teleoperation with time delay	ICRA 1990 [S]	predict the robot's motion over the delay for the operator
IEEE 802.11e (WMM)	standard	access categories for priority traffic

4. Where the Gaps Are

Table 11. The properties of the scenario against the nearest prior work.

Property	Nearest prior	What is missing
errors compound, and the link is one of them	localisation (ATE); grasp detection (success on fixed arms); offloading (average accuracy)	one error budget from map to fingertip in which delay is converted to distance at the gripper
tolerance and delay budget shrink together	FogROS2-LS latency targets; coarse-then-fine grasping	stage-wise delay budgets derived from tolerance and speed, and the measured point where offloading stops paying
where a stage runs is a choice	Chinchali's offloading policy; Neurosurgeon's cut; Glimpse	a split that moves with the motion stage, scored at the grasp rather than per query
the robot can predict part of the delay	Smith predictor; predictive displays in teleoperation	motion-compensated answers from a remote perception model, combined with speed adaptation for the unpredictable remainder
the Cartesian arm needs the base	whole-body servoing on arms with lateral joints; OK-Robot open-loop grasping	an on-robot base-yaw and arm-extension servo for the stages no round trip can serve, with the server reduced to proposals and verification

In short: offloading and video-analytics work optimises what crosses the link without an actuator at the other end; manipulation work optimises the grasp without a link in the loop; systems on Stretch use a remote workstation and report overall success. Converting delay into gripper error, cancelling the part the robot can predict and slowing for the rest, closes the space between them. Chinchali et al., DDS and OK-Robot are the three to position against first.

5. What Is Asked of the Collaborators

- The Stretch 4 configuration: lidar (planar or 3-D), head and wrist cameras, gripper, on-board compute, ROS 2 distribution.
- The current pipeline: which models run on the server, their inference time and what they return; whether the grasp is open-loop from the head camera; which localisation runs and in which mode.
- The object set, the surfaces and the success criterion.
- Permission to run chrony and a logger on robot and server, to throttle the robot's link for experiments, and to mount AprilTags; three sessions of about two hours.